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Linking knowledge, attitudes, and practices to disease dynamics in tilapia (*Oreochromis niloticus*) aquaculture on Volta Lake

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Introduction: Tilapia (Oreochromis niloticus) aquaculture accounts for nearly 80% of Ghana's farmed fish production, with cage culture on Volta Lake as the dominant system. In recent years, production has been severely constrained by high disease-related mortalities, notably due to Streptococcus agalactiae and Infectious Spleen and Kidney Necrosis Virus (ISKNV). This study investigated how farm management practices and farmers' knowledge, attitudes, and perceptions influence disease dynamics on Volta Lake.

Methods: A cross-sectional survey of 56 farms across five districts was conducted using semi-structured questionnaires and interviews with farmers and the local fish health officer. Results showed that disease outbreaks were reported in all farms, with mortalities ranging from 10-80%.

Results: Farmers attributed outbreaks primarily to pathogens, but also to poor water quality, high stocking densities, and inadequate biosecurity. Diagnostic practices varied, with most farmers relying on clinical signs or peer consultation rather than laboratory testing. Clinical signs commonly reported included exophthalmia, swollen abdomens, and skin lesions. Farm management practices such as grading, sourcing of fingerlings, water quality monitoring, and biosecurity were inconsistent and largely unstandardized. Regression analysis indicated that stocking density, biosecurity, and water quality monitoring did not significantly predict disease-related mortality, underscoring the multifactorial nature of outbreaks in this open water system.

Discussion: The findings highlight the need for coordinated extension services, accessible diagnostic facilities, and standardized farm management protocols to strengthen disease prevention and control in Ghana's cage aquaculture sector.

KEYWORDS

KAP (knowledge, attitude and practice) study, disease management, Ghana, Lake Volta, tilapia farming

1 Introduction

Tilapia culture accounts for approximately 80% of Ghana's total aquaculture production. (Amenyogbe et al., 2018; Aheto et al., 2019) with cage culture industry suppling majority of this production output on the Volta Lake (Mensah et al., 2018). The demand for tilapia is high, presenting an opportunity for the industry to expand and optimize its operations. However, tilapia production in the country has declined due to increased disease problems and high mortality rates (exceeding 50%), negatively impacting the economic returns on investments (Ramírez-Paredes et al., 2021). This has led to a decrease in the number of operational cages on the lake. The infectious spleen and kidney necrosis virus (ISKNV) disease outbreaks on the Volta Lake for instance, has had a devastating impact on Ghana's aquaculture industry since its emergence in late 2018, particularly affecting tilapia under intensive culture. The virus, which primarily targets juvenile fish, has led to mortality rates ranging from 60% to 90% in some intensive cage culture systems with severe economic repercussions (Ramírez-Paredes et al., 2021). Survival rates at the grow-out stage have been reported to be as low as 5–20%, with episodic mortality outbreaks occurring shortly after stress events such as sex reversal or translocation to lake cages (Ayiku et al., 2024).

Research on the causes of diseases in the early 20th century mainly focused on pathogens as the cause of infections. However, recent studies and developments have revealed the importance of viewing diseases as the result of multiple interacting factors (Zornu et al., 2023). Fish health can be understood as the interface between the fish, the environment, and pathogens. A healthy fish will hardly succumb to infections, provided it is under favourable culture conditions (Roberts, 2012), and management practices can hence play a crucial role in disease occurrence. For instance, high stocking densities in intensive cage culture, aimed at increasing profits, can lead to poor water quality and suboptimal feeding regimes, and high incidence of physical injuries exposing fish to stress levels that compromise their health and well-being. Work done by Huchzermeyer (Huchzermeyer, 2018), indicates that under high density conditions, the competition for food often results in injury to the eyes and fins of cultured fish and this predisposes them to secondary infection.

Studies investigating the causes of diseases have primarily focused on the pathogenic causes with a few examining the environmental aspects of disease occurrence. A recent study (Abarike et al., 2022) on fish health management practices in Volta Lake revealed that farmers believe a combination of pathogens and poor water quality contributes to the incidence of diseases on their farms. Additionally, 10 of the 30 farmers interviewed on Volta Lake regarding the 2018 ISKNV disease outbreak reported observing changes in water color and recorded water quality parameters before the outbreak (Ayiku et al., 2024). Others reported that diseases in the lake are caused by the interaction of pathogenic and non-pathogenic factors, and they identified some farm management practices as risk factors for disease occurrences (Zornu et al., 2023). However, more evidence is needed to ascertain these research claims and aim policy directives for effective disease management on the lake. Thus, this study aimed to provide more information on the farm management practices on the lake and how they influence disease occurrence and spread within the lake.

2 Methodology

2.1 Study area

The study was conducted on the Volta Lake of Ghana, within stratum II. The lake has an average depth of 19 meters and an area of 8,502 square kilometers. This area was selected because it is the primary location for cage culture on the lake. The study focused on five districts that belong to three administrative regions around the lake. These were the Asuogyaman and Lower Manya Krobo districts of the Eastern Region, the Afadjato South and South Tong districts of the Volta Region, and the Shai Osudoku district in the Greater Accra Region.

2.2 Selection of respondents and respondent characteristics

The selection of farms for this study was mainly based on the availability and willingness of farms to participate in the research. First, a list of farms (196) on the lake and their contact details was obtained from the secretary of the Ghana Aquaculture Association at the time of data collection (January 2024), who doubles as an Aquaculture extensionist for farmers on the lake. The farms were then contacted through phone calls, and their consent and interest in the research were sought verbally before proceeding with interviews. Initial contact with the farms was made in December 2023, and a reminder call was made one day before the survey. To be considered as an unavailable or non-response farm, the farm (contact called) had to explicitly declare non-interest in the study, or the contact was unreachable for all the times the calls were made. Farms were considered non-reachable/non-inclusive after no response or non-connectivity after being called twice a week for 3 weeks. These initial calls resulted in 38 positive responses for the farm visit. However, in some of the study districts, such as Afadjato South, Asougyaman, and South Tong, initially identified farms referred other farms (16 more) that were unreachable or non-responsive to participate in the study through snowball sampling. Thus, a total of 56 farmers were finally interviewed in this research, in addition to the fish health officer in charge of the study area. Non-participation of farmers was mainly due to the unavailability of farmers during the study period, concerns regarding data confidentiality, or logistical constraints related to the accessibility of the farm. Selected farms varied in production capacity, representing the general lake productivity. Thus, the farms interviewed included all production levels: small (58.9%), medium (25%), and large-scale (16.1%) production. Non-respondent and non-inclusive farms also fall within these production criteria. Non-response in this study did not receive any adjustments because the focus of the study was exploratory in nature.

Aside from the Veterinary officer for the area, all the people interviewed on the farms held different positions and had various responsibilities on the farms. Some of these individuals were farm owners, owners who also served as farm managers, farm managers themselves, or farm workers or employees. Additionally, approximately 25 respondents were individuals who acquired the skills and practices of aquaculture through their work on fish farms, thereby undergoing on-the-job training. Twenty-three (23) respondents from the farms and the veterinary officer had received aquaculture training through the study of aquaculture programs in their tertiary education or short aquaculture training programs, with or without an acquired certificate. In contrast, the remaining eight (8) respondents had no formal aquaculture training or on-the-job training of any form.

2.3 Study design and data collection

The study was conducted as a cross-sectional study from January 16 to 19, 2024. Data was collected using a semi-structured questionnaire after a pretest. The final version was drafted following comments and feedback from the pretest. Paper or electronic (Kobo Toolbox Software) versions of the questionnaires were administered orally and face-to-face to the farm respondents and electronically for the veterinary officer. The questions focused on the general farm characteristics, fish disease history, suspected causal agents and/or factors and farm management practices. Respondents were asked to recall the estimated fish mortalities for the production cycle during outbreaks of disease. Information sought from fish health officer was on existing regulations and adherence regarding fish biosecurity and fish health management, as well as disease history records and the causes of diseases.

2.4 Recording of data

Disease outbreak information was collected as the year of first reported outbreak within the study window (2013–2023), with missing values indicating that respondents did not know the year. Where available, respondents consulted farm records; otherwise, responses relied on recall. In addition, respondents reported whether outbreaks were associated with adult fish mortality (yes/no). Mortality was therefore measured at the farm level and could not be weighted by stock at risk because contemporaneous stock-at-risk data at the time of the first outbreak were not available. Because all farms reported disease occurrence, there was no variance in the binary "disease occurrence" outcome and sensitivity analyses for occurrence were not informative. Sensitivity analyses were instead conducted for outcomes with variability, i.e., mortality (yes/no), by comparing unweighted summaries with stock-proxy-weighted summaries (using reported stocked fish at interview as a proxy for stock at risk) and by applying trimmed weights to assess the influence of extreme values.

Adult fish mortality was analyzed as a binary farm-level outcome (Fish death Stage_Adult: yes/no). We used a generalized linear model (binomial family with logit link). Analyses were conducted at the farm level using complete cases for the outcome and included predictors. Multicollinearity was assessed using variance inflation factors (VIF). Model adequacy was evaluated using residual diagnostics (binned residual plot and deviance residuals versus fitted values). To account for within-district clustering, we reported district-clustered robust standard errors (CR2); given the small number of districts, clustered inference was treated as a sensitivity check.

2.5 Data analysis

Questionnaire data were entered from paper forms or exported from Kobo Toolbox into Microsoft Excel (Microsoft 365) for cleaning and descriptive summaries (tables, graphs). Statistical analyses were conducted in R and Stata 18. Descriptive statistics were produced for all variables, and candidate predictors were screened to avoid inclusion of near-constant variables (very high proportions of 0 or 1), except where required for disease reporting variables.

To assess associations between management practices and adult fish mortality, we fitted a generalized linear model (logistic regression; binomial family, logit link) with adult mortality recorded at farm level (yes/no) as the outcome. Given clustering by district and the small number of districts, inference used district-clustered robust standard errors (CR2) with Satterthwaite-adjusted tests. Multicollinearity was assessed using variance inflation factors (VIF), and model adequacy was evaluated using residual diagnostics (including binned residual plots and deviance residuals vs fitted values). Missing data were handled using complete-case analysis for the variables included in each model. Statistical significance was assessed at $\alpha = 0.05$.

Sensitivity models explored (i) adjustment for a farm-size proxy (log-transformed reported number of fish stocked at interview), and (ii) inclusion of district effects; these sensitivity analyses were interpreted cautiously due to small sample size and potential separation/instability when adding multiple district indicators.

Overall, KAP constructs were operationalized as a combination of (i) single-item measures (e.g., overall aquaculture knowledge rating) and (ii) pre-specified composite indicators based on multi-response checklists. Binary items were coded 1="Yes/endorsed/performed" and 0="No," with higher composite scores indicating greater intensity of the construct (e.g., more monitoring activities or more sharing pathways). Because many practice measures represent formative checklists (i.e., distinct actions that need not co-occur), internal consistency metrics were not assumed to be appropriate for all composites and were only estimated for small sets of items intended to reflect a common response dimension (e.g., parameter-change checklists). To avoid unstable inference from quasi-invariant predictors in small samples, candidate predictors were screened for near-constancy (<5% or >95% prevalence of "1" among analyzable farms) and such variables were excluded from

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multivariable models, except where required for descriptive reporting of disease observations. Aggregation rules for each KAP subscale items included scoring (Supplementary Table 1), along with the full questionnaire (Supplementary File S1).

Mortality modelling. To assess associations between management practices and adult fish mortality, mortality was analyzed as a count outcome using a negative binomial regression framework to accommodate overdispersion. The number of fish stocked (recorded during the interview and used as a proxy for stock at risk) was included as an exposure term, such that model coefficients represent incidence rate ratios (IRR) for mortality per stocked fish. Farms categorized as non-reporting were excluded to avoid bias from misclassifying missing outcomes as true zeros. Robust standard errors were used, and a sensitivity specification included district fixed effects to account for district-level heterogeneity.

3 Results

3.1 Selection of respondents and Respondent characteristics

Fifty-six (56) out of a total of 196 identified farmers (Figure 1) were interviewed in this research, in addition to the fish health officer in charge of the study area.

3.2 Farm characteristics: type of facility, cage type and sex of cultured fish

Different farms operate one or multiple facilities, including hatcheries, nurseries, or grow-out facilities. About 58.9% of the farms operated only grow-out facilities, 1.9% operated only a hatchery, and 7.1% operated both a nursery and a grow-out facility. In contrast, the remaining 32.1% had all three facilities. The most used cage type identified in this study was the square cage type (5x5x5 and 6x6x6), as 96.4% of the respondents used this. The study focused on tilapia culture on the lake and identified 25% of the farms to be practicing mono-sex culture while the remaining 75% practiced a mixed-sex culture.

A substantial minority of farms reported having hatchery facilities (19/50; 38% among farms with non-missing responses). Information on fingerling sourcing was incomplete (22/56 missing) and, where reported, indicated multiple sourcing pathways, limiting inference on the proportion sourced from accredited hatcheries versus other farms.

3.3 Fish disease occurrence and history

All the farms interviewed indicated they had experienced and still encounter diseases in their production activities for all fish sizes, with the

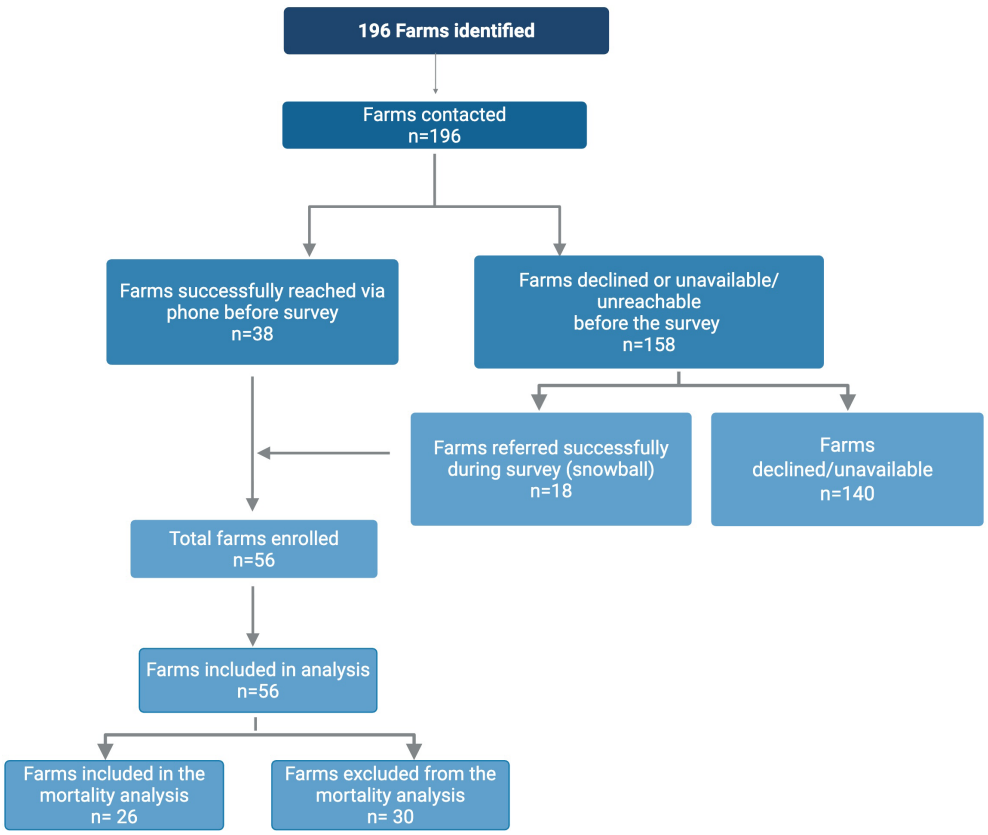


FIGURE 1
Flow diagram showing farm selection and inclusion in analysis (created in https://BioRender.com).

first occurrence reported in 2013, the last in 2023 (year of interview). The frequency of disease occurrence varied among farmers as some reported disease to occur annually. For others, diseases are seasonal or occur once every quarter, and some cannot even specify the exact times of disease occurrence, summarized in Figure 2.

Additionally, regarding the cause of disease, farmers are of the view that diseases are primarily caused by infectious agents, such as Streptococcus agalactiae, Infectious Spleen and Kidney Necrosis Virus (ISKNV), and others, including Aeromonas spp. and fungi (Figure 3).

According to the fish health officer in charge of aquaculture on the lake, they record diseases every month, mainly caused by *Streptococcus agalactiae*, Infectious Spleen and Kidney Necrosis Virus (ISKNV), and *Aeromonas* spp. S. agalactiae and ISKNV are the most prevalent.

A closer look at the two primary pathogens, *S. agalactiae* (Figure 4) and ISKNV (Figure 5) of infectious diseases across the various districts under this study reveals that Afadjato South had about 84% of its respondents attributing the diseases to *S. agalactiae* infection (Figure 4). In contrast, farmers in Shai Osudoku reported no infections by *S. agalactiae*. On the other hand, infections caused by ISKNV were the least reported by farmers in the Lower Manya Krobo district, with the highest number being in the Afadjato South district (Figure 5).

Responses from the farmers indicated they believed the causes of diseases were more a result of an interaction between two or more infectious agents, as most farms reported more than one pathogen as the cause of infection. Apart from pathogen attacks, both farm respondents and veterinary officer also reported factors such as inadequate biosecurity measures, poor water quality, seasonal variation, high stocking density, and poor handling as contributing factors to disease occurrence within their farms, with poor water quality being the highest contributing factor (Figure 6).

3.4 Origin of diseases

According to farmers and farm managers interviewed, neighboring farms have also experienced disease outbreaks. While half of the respondents (50%) could not identify the origin of diseases, 44.6% confirmed that diseases originate from farms upstream, and the remaining 5.4% presumed that the origin of diseases is from farms downstream. This finding was same response from the fish health office, which says that common areas on the lake with frequent disease outbreaks are the upstream and downstream sections.

3.5 Diagnosis of disease

Diagnostic methods and routines are not well aligned and vary across districts (Figure 7). Very few farms interviewed took samples to the laboratory or sought the expertise of extension service officers who diagnose disease through postmortem examination and other laboratory procedures. Some farms also combined two or three of these means of diagnosing diseases. While taking samples to the laboratory for diagnostic follow-up was reported only in the Lower Manya Krobo district, respondents in Afadjato South and Asuogyaman, primarily diagnosed diseases on their farms by consulting other farmers based on their observed clinical signs. For the most part, diagnosis is based on observing clinical signs.

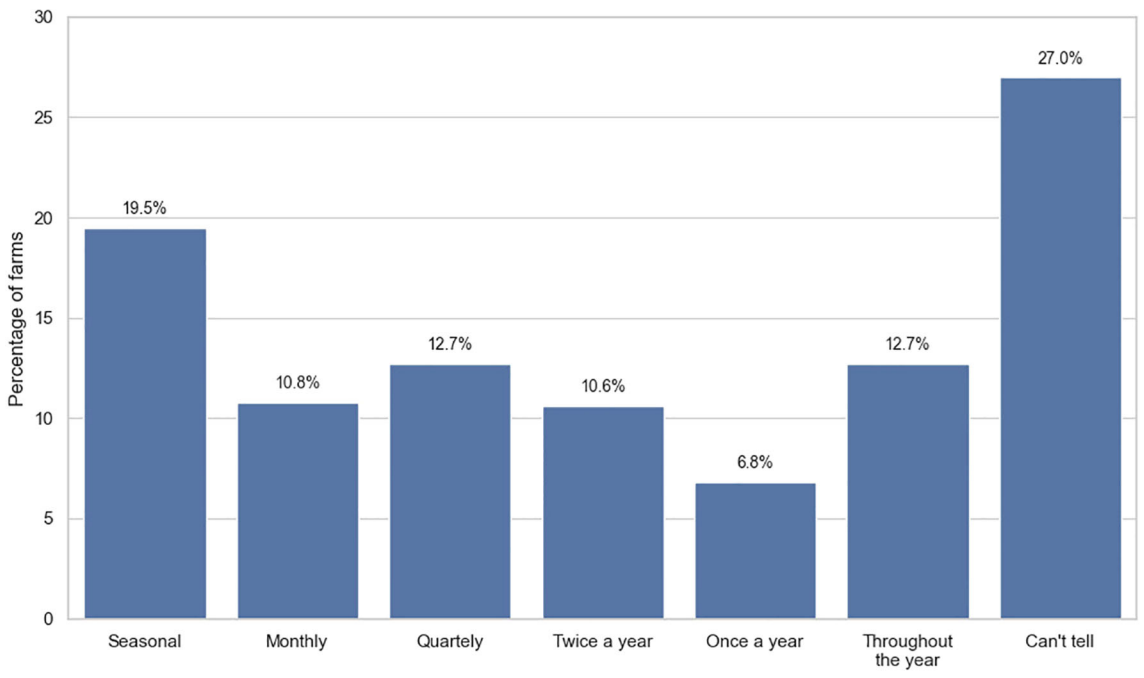


FIGURE 2
Frequency of disease occurrence in the farms.

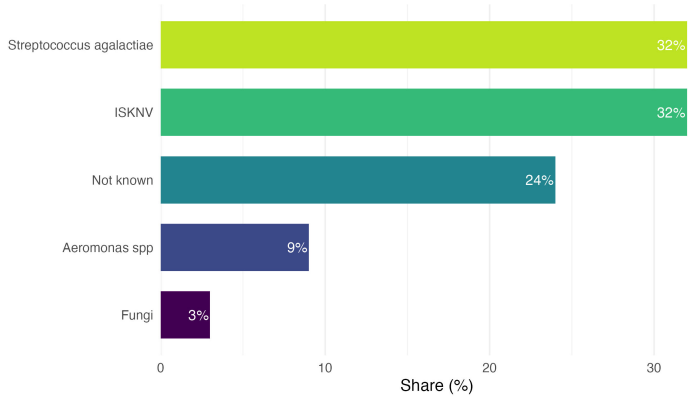


FIGURE 3
Infectious agents considered by the farmers to cause diseases on the Volta Lake.

Pathogen attribution was primarily based on farmer assessment rather than laboratory confirmation. Among farms with non-missing responses, only 3/51 (5.9%) reported taking samples to a laboratory, whereas 52/54 (96.3%) reported relying on clinical signs; 23/51 (45.1%) reported consulting other farmers, and 5/52 (9.6%) reported contacting extension officers for diagnosis. Use of laboratory testing was uneven across subgroups: it occurred in only two districts (Asuogyaman and Lower Manya Krobo) and was reported only among respondents in the highest recorded education category in the dataset.

Another factor that influenced the diagnosis of diseases by farmers as either of Streptococcus agalactiae origin or ISKNV was the educational background of the farmers or farm managers. For *Streptococcus agalactiae*, educational background (formal or no

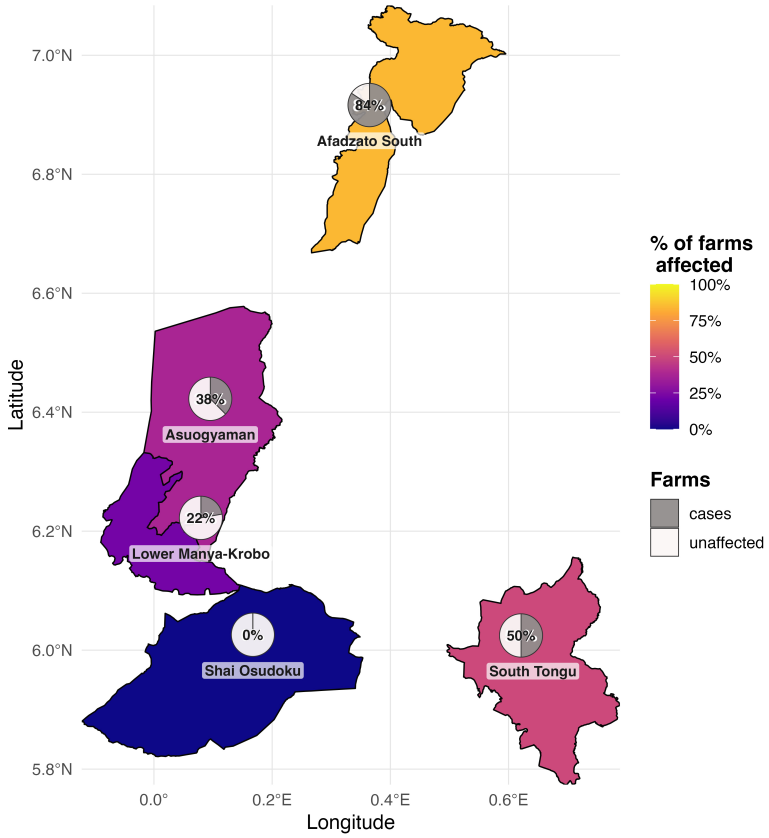


FIGURE 4
Reported incidences of *Streptococcus agalactiae* infection in the districts studied.

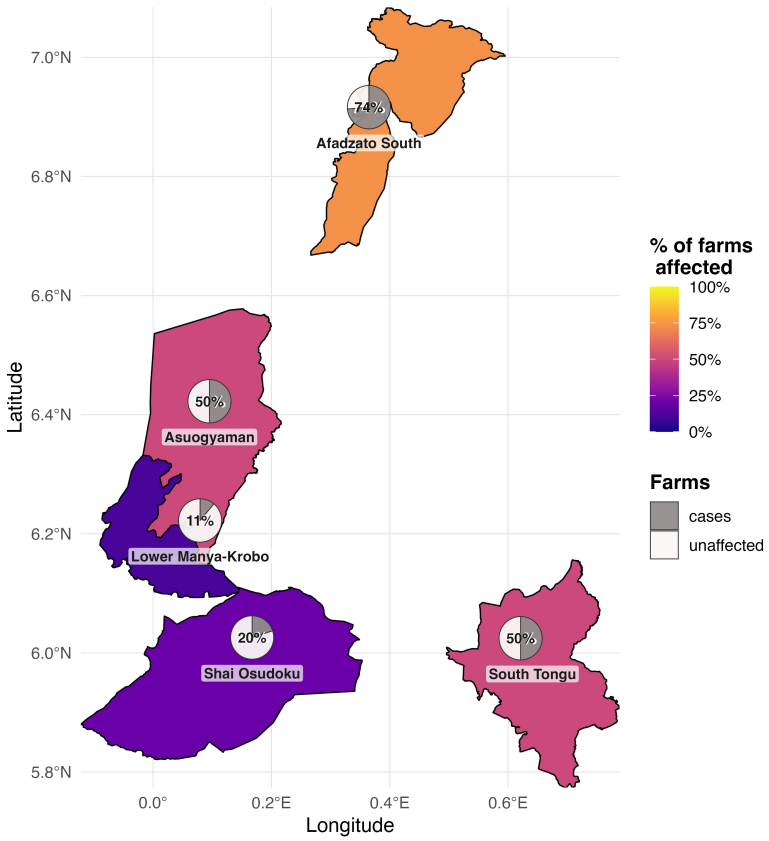


FIGURE 5
Reported incidences of ISKNV infection in the districts studied.

formal education) did not significantly influence the attribution of infection to the bacteria (p = 0.0923, Figure 8a). However, with ISKNV, respondents' educational background or level of education influenced the attribution of the disease to the virus. Respondents with secondary or tertiary education were more likely to report ISKNV infections on their farms (p=0.0139, Figure 8b). Poor handling was also found to be a significant predictor for the occurrence of ISKNV and *Streptococcus agalactiae* (Figure 9).

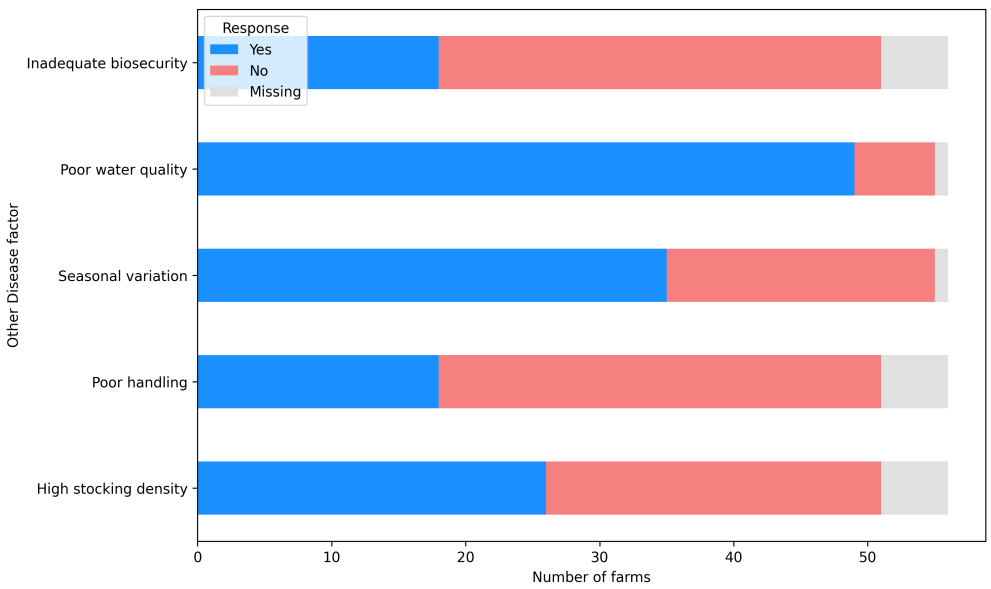


FIGURE 6
Other factors contributing to the occurrence of diseases on the Volta Lake.

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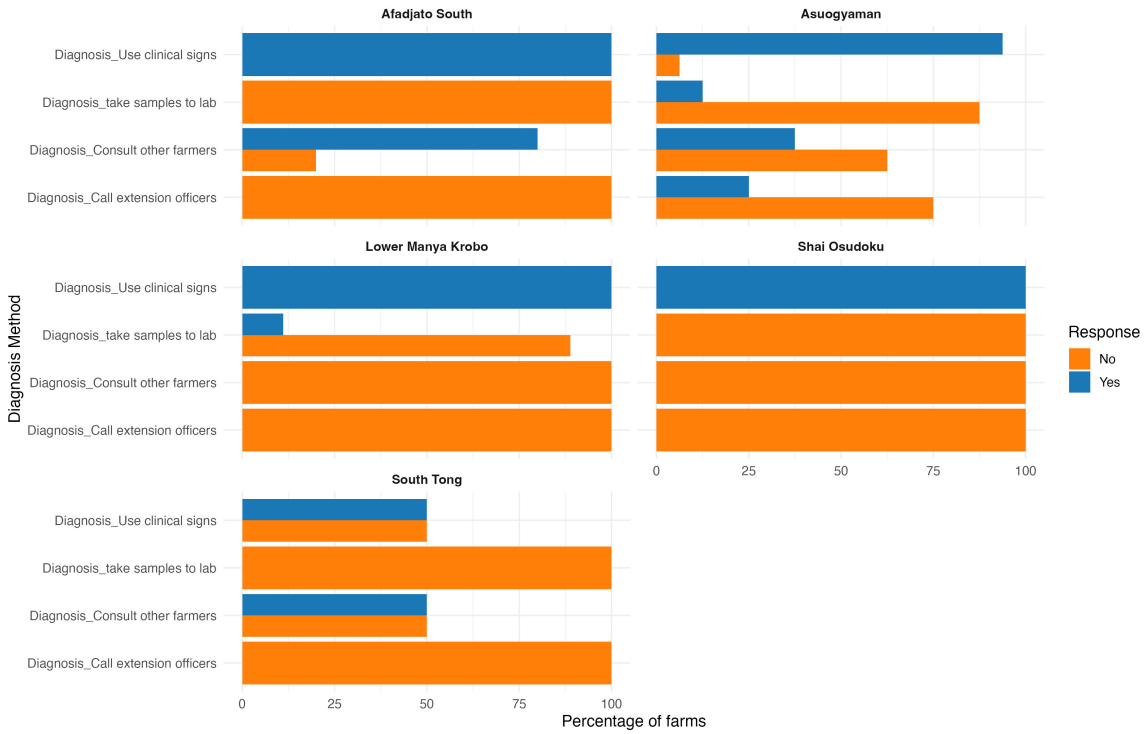


FIGURE 7
Means of diagnosing fish diseases on the farms on the Volta Lake by the study districts.

3.6 Common clinical signs of disease on farms

Diseased fish showed either one or combinations of such clinical signs as decreased feeding, poor growth, stationery at one place, abnormal swimming, skin lesions/hemorrhages, change in skin color, bulging eye, swollen abdomen, inflammation of internal organs, pale gills, fin rot, and cottony growth on the mouth (whitish mouth). All interviewed farms have fish death or high mortality as the major disease sign. Mortality rates ranged from as low as 10% of the fish stocked to as high as 80%. These reported ranges were based on farmers' estimations of deaths as a proportion of total fish stocked. According to the veterinary officer in charge of the study area, the above signs are common indicators of diseases in the lake. Aside from fish mortality in all the studied districts, exophthalmia (bulging eyes) and swollen abdomen were among the most common signs farmers observed and reported (Figure 10).

3.7 Source of fingerlings and stocking density

Interviewed farmers disclosed that their fingerlings were sourced from farm-owned hatcheries, other farms or government-accredited hatcheries. Thirteen (13) farms produced their fingerlings, 19 sourced them from different farms around them, and only two (2) farms sourced from the government facility. At the same time, the remaining 22 did not disclose the source of their fingerlings. Of the farms that sourced their fingerlings from accredited national hatcheries, only one confirmed the acquisition of the certification of the health status of stock. About 63% of those that procured fingerlings from other farms and 46% of self-owned confirmed certification of fish stock as disease-free. The certification of fish stock as disease-free was mostly verbal as most farms did not have any documentation to prove that. Formal health certificates accompanying fingerling consignments were not consistently available and were not captured as a structured variable in this survey; instead, acquisition of health information on fingerlings and conduct of personal health assessment were recorded as proxies for pre-stocking health assurance.

3.8 Farm management practices

3.8.1 Grading

General farm management practices among fish farmers on the lake included fish grading, which was employed by 62.5% of the fish farms studied. The grading time varied among the farms, with some receiving multiple grades within the production cycle. While 62.9% of these farmers do grading before stocking, 2.9% do grading both before stocking and two weeks after stocking, 17.1% grade after one month of stocking, 5.7% do grading at every sampling time, another 5.7% grade their fish two months after stocking and the remaining 2.9% of the farms that practice grading does so after three months of stocking.

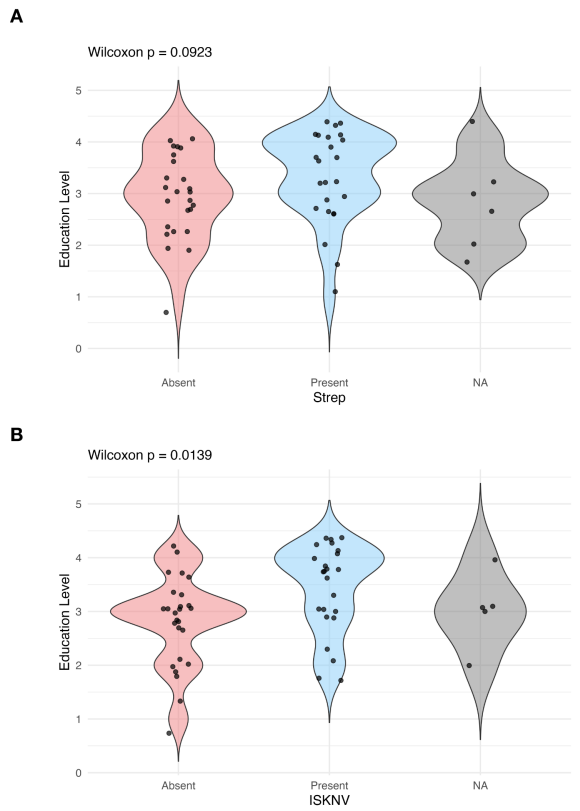


FIGURE 8
Influence of the educational level of respondents on the occurrence of **(A)** *Streptococcus agalactiae* or **(B)** ISKNV infection. Educational levels: 1, No formal education; 2, Primary education; 3, Secondary education; 4, Tertiary education. Wilcoxon's test was used to compared 'Absent' and 'Present', and p-values are shown.

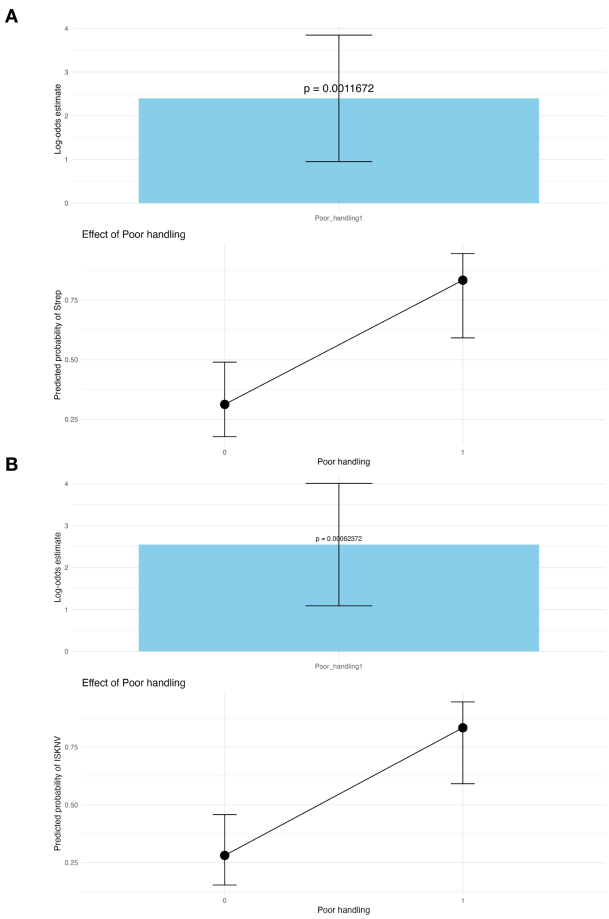


FIGURE 9
Poor handling as a predictor for **(A)** *Streptococcus agalactiae* or **(B)** ISKNV infection. For both, poor handling comes out as a significant predictor.

3.8.2 Biosecurity measures of farms

Biosecurity measures implemented by farmers or farm managers in this study included obtaining disease-free stock, disinfecting people and tools, and restricting movement to specific areas of the farm. To ensure the stock is disease-free, different farmers employed various methods, including acquiring health status information on the fingerling or fish stock from hatcheries, conducting personal health assessments on the stock, and implementing quarantine measures.

3.8.3 Water quality monitoring

Monitoring of the water quality by the farmers was found to be relatively uncommon in the surveyed cage fish farms. Approximately 73.2% of respondents reported having no monitoring system in place. According to the farmers, the main challenges in monitoring water quality included insufficient access to relevant laboratories and equipment, as well as financial constraints. Moreover, some farmers perceived that, since the lake is an extensive open system, monitoring water quality on a per-farm basis would not impact the overall water quality of the lake. The frequency of monitoring for those who did monitoring varied between daily, weekly, biweekly and monthly tracking and monitoring when there was a problem on the farm. Though most farms did not monitor water quality, most (those who measured and those who did not) acknowledged a change in some water characteristics, mainly before and during disease outbreaks. According to the farms, the most frequently reported changes were changes in water color (water turns darker), temperature, and dissolved oxygen levels. Water quality monitoring for farms that measured was by using water quality probes and laboratory analysis following standard procedures.

3.8.4 Treatment or control measures for diseases

According to the fish health expert in charge of the lake, the best treatment advice they give farmers in cases of diseases is the implementation of biosecurity measures and good farm management practices. However, they acknowledge farmers' use of human antibiotic drugs in treating diseases on the lake, although they do not recommend this practice. Eighteen of the respondents from the farms also reported that their main treatment options for diseases, which included withdrawing feeding (61.1%), treating with potassium permanganate (5.6%), or performing heat shock treatment (11.1%) before stocking fish. Other treatments or control measures included signing up for the government

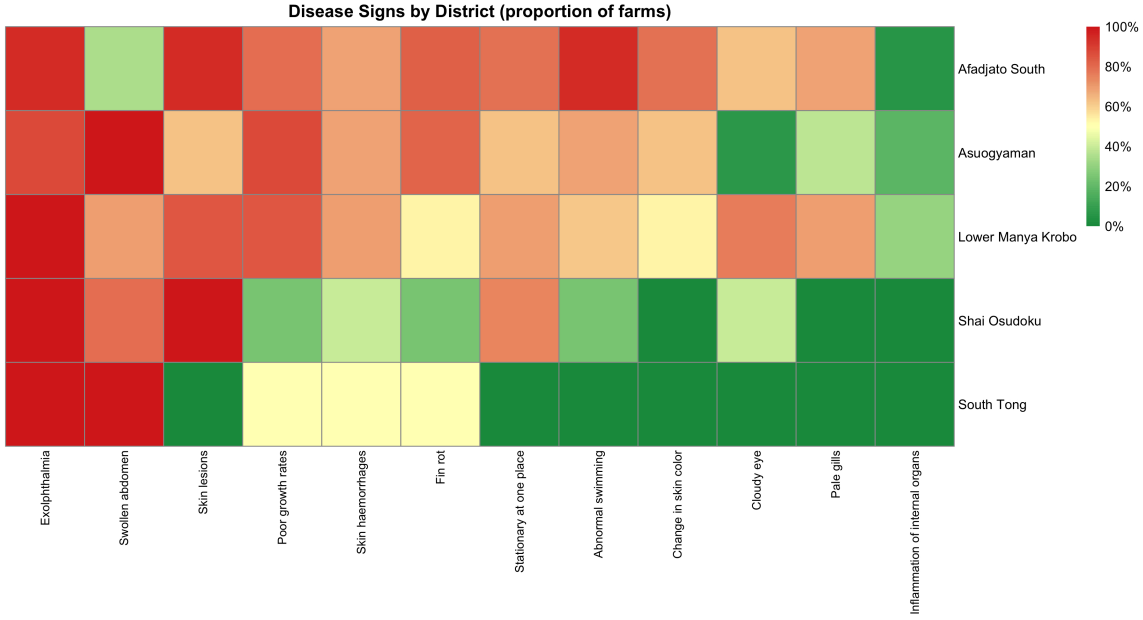


FIGURE 10
Clinical signs recorded by farmers in the various districts.

vaccination program (11.11%) and using the commercial probiotic consortia, Rhodomax (5.6%), to modulate the intestinal ecology of fishes for enhanced survival and growth. Vaccination programs (Irido vaccine) for disease management on the lake were introduced by the government of Ghana during the ISKNV disease outbreak in the latter part of the year 2019.

3.9 Effects of farm management practices on disease-related fish death

Different farms stocked their fish to varying densities with different cage dimensions. The standard dimensions of cages used by fish farmers on the lake were as small as 21.25m3 (2.5x5x1.7). The most commonly used cages were the 6x6x6 (216 m^3) cages, constituting 33.9% of the farms, followed by the 5x5x5 (125 m^3) cages, accounting for 30.4%. Some farms also used cages as large as 320 m^3 (8x8x5). The number of fish stocked per cubic meter varied considerably among the different farms. This was obtained using the cage dimension and total number of fish in a cage provided by the farms, as some could not provide the stocking density outrightly. The stocking density obtained ranged from as low as 7 fish/m^3 to as > 130 fish/m^3, while some farms did not disclose information on stocking density (and are excluded).

In negative binomial rate models of reported mortality counts with the number of fish stocked included as an exposure term, grading of fish and between-farm sharing were not jointly associated with mortality rate (Wald $\chi^2(2)=2.63$, $p=0.268$). Derived stocking density (fish/m^3), calculated from reported fish stocked and cage volume, was not associated with mortality rate when modelled continuously (per +10 fish/m^3), on the log scale (Table 1). Table 1 summarizes the negative binomial mortality rate model using reported adult mortality counts as the outcome and the number of fish stocked as the exposure term (rate per stocked fish). In this model (n=26), neither grading of fish (IRR 0.80, 95% CI 0.61–1.05, p=0.112), between-farm sharing (IRR 1.08, 95% CI 0.72–1.61, p=0.712), nor calculated stocking density (per +10 fish/m^3; IRR 1.07, 95% CI 0.88–1.29, p=0.512) was associated with mortality rate. Overdispersion supported the negative binomial specification (alpha ≈ 0.25; LR test of alpha=0, p<0.001). A sensitivity model including district fixed effects did not materially improve fit (LR $\chi^2(4)=6.25$, $p=0.181$). Estimates were imprecise due to the modest analytic sample size and potential measurement error in derived density (self-reported cage dimensions and stocking numbers recorded). Next (Table 2), density was categorized into bands reflecting low (≤20), moderate (20–50), high (50–130), and very high (>130) densities (n=26). None of the density categories differed from the reference group (≤20 fish/m^3), and the density block was jointly non-significant (Wald $\chi^2(3)=0.92$, $p=0.822$). Across these specifications, grading and between-farm sharing remained jointly non-significant (Wald $\chi^2(2)=2.63$, $p=0.268$), indicating that conclusions were robust to the density parameterization.

TABLE 1 Negative binomial mortality rate model for adult fish mortality (reported deaths) with fish stocked as exposure.

Predictor	IRR	95% CI	P-value
Grading_of_fish	0.80	0.61–1.05	0.112
share_between_any	1.08	0.72–1.61	0.712
ln(dens)	1.07	0.88–1.29	0.512

Incidence rate ratios (IRR) are reported with 95% confidence intervals; density is scaled per +10 fish/m^3.

TABLE 2 Sensitivity analyses for stocking density (fish/m^3) in negative binomial mortality rate models: log-transformed density and categorical density bands (≤20, 20–50, 50–130, >130 fish/m^3).

Predictor	IRR	95% CI	P-value
Grading_of_fish	0.81	0.62–1.05	0.112
share_between_any	1.10	0.71–1.71	0.672
dens_cat: 20–50 (ref ≤20)	1.15	0.71–1.88	0.565
dens_cat: 50–130 (ref ≤20)	1.18	0.73–1.90	0.500
dens_cat: >130 (ref ≤20)	1.08	0.63–1.86	0.785

IRR with 95% confidence intervals is reported.

4 Discussion

4.1 Disease history and causes

Diseases remain a significant concern for tilapia farmers on Volta Lake, affecting their economic returns. All the farmers interviewed in this study confirmed the presence of diseases on their farms, which was an all-time event on their farms. According to farmers and veterinary personnel, the primary cause of diseases is mainly pathogenic, with pathogens including bacteria, viruses, fungi, or sometimes a combination of these. Streptococcus agalactiae and the infectious spleen and kidney necrosis virus were the most mentioned as being responsible for diseases on the lake which conform to the findings from previous studies (Verner-Jeffreys et al., 2018; Ramírez-Paredes et al., 2021; Zornu et al., 2023; Duodu et al., 2024). Apart from the pathogenic influence of disease occurrence on the lake, fish farmers and the veterinary office mentioned that water quality, resulting from water pollution caused by agricultural farms and waste discharges from industrial effluents, poses significant stress on the fish, making them more susceptible to pathogens already present in the lake. Additionally, according to farmers, the changes in water quality observed before and around the time of disease occurrence were attributed to seasonal variations in weather conditions, which altered water temperature. This finding confirms similar findings from earlier studies (Abarike et al., 2022; Zornu et al., 2023), where poor water quality was reported as a major contributing factor to diseases on Volta Lake.

4.2 Diagnosis of diseases

Disease diagnosis was primarily conducted by a combination of methods or, in some cases, by consultation with nearby farms following the observation of clinical signs. Only a few farms in only one of the districts studied consulted extension services (veterinary officers) or did actual laboratory examinations. According to farmers, this was due to financial constraints associated with sending samples to the laboratory, the unavailability of extension services, and sometimes the accessibility of these services. On the part of the veterinary office, the non-compliance of these fish farmers with biosecurity and fish farming regulations prevents them from approaching the office for help in a timely manner. Lack of enforcement and non-compliance of regulations by farmers in the study area has been reported earlier (Zornu et al., 2023). Another factor that influenced the diagnosis of diseases in this study was the educational level or background of the respondents. ISKNV was reported by respondents with relatively higher educational backgrounds compared to S. agalactiae, which was not linked to the educational background. This could be explained by the fact that S. agalactiae infection has been present in the lake for some time, so farmers may have been aware of its presence as well as the signs of that infection, as opposed to ISKNV, which is a more recent finding in the lake. Unlike other districts, farmers in the Afadjato South District were able to attribute disease outbreaks to specific pathogens. This was because they relied on social interaction within the fish farming community, which utilized clinical signs to diagnose diseases. Thus, they relied on more on the experience of farmers that have had those disease conditions on their farms previously.

4.3 Clinical signs of diseases on the Volta Lake

The clinical signs observed in diseased Nile tilapia on Volta Lake across the studied districts ranged from reduced feeding and abnormal swimming behavior to skin lesions and systemic signs, including exophthalmia, swollen abdomen (abdominal distension), and internal organ inflammation. These signs are indicative of severe underlying infectious conditions. The widespread reporting of mortality, with rates ranging from 10 to 80%, highlights the significant impact of diseases on aquaculture operations in the study area and the country, as the lake supplies approximately 80% of the national aquaculture production output. These findings are consistent with those previously reported for major bacterial and viral pathogens affecting tilapia culture in the study area and globally. For instance, farmers interviewed on the Lake experienced disease-related mortalities ranging from 1% to 95%, with an average of 63% (Duodu et al., 2022). The high incidence of clinical signs, such as exophthalmia and abdominal distension, reported by farmers suggests that systemic infections are likely caused by Streptococcus agalactiae or infectious spleen and kidney necrosis virus (ISKNV), as both pathogens are known to produce such pathognomonic features (Duodu et al., 2022). S. agalactiae is an important pathogen in tilapia culture systems and infections are characterized by a distended abdomen, opaque eyes, skin hemorrhage, exophthalmia, and high mortality (Duodu et al., 2022; Mohamed et al., 2023). Similarly, ISKNV, which is a member of the Megalocytivirus genus, has emerged as a significant viral pathogen in tilapia culture, inducing such clinical signs as exophthalmia, lethargy, change in skin coloration and widespread tissue necrosis (Ramírez-Paredes et al., 2021; Duodu et al., 2022; Ayiku et al., 2024). In addition, other signs, such as hemorrhages, ulcers, fin rot, and pale gills, indicate a likelihood of opportunistic infections by *Aeromonas* spp., although most farmers do not commonly report this. These bacterial pathogens have been

reported in some studies of tilapia diseases on the Volta Lake, in association with some of the clinical signs observed in this study (Abarike et al., 2022; Duodu et al., 2022). Due to their opportunistic nature, these signs may occur as secondary infections following immunosuppression caused by a primary viral or bacterial pathogen (Duodu et al., 2022), where about 63% of the Aeromonas-positive samples also tested positive for ISKNV.

Most importantly, the overlap of clinical signs for the various pathogens reported by fish farmers in this study presupposes a possible co-infection of these pathogens and others not stated in this study. This has been previously hinted at by several authors who have conducted epidemiological studies on the lake (Ramírez-Paredes et al., 2021; Zornu et al., 2023; Ayiku et al., 2024; Duodu et al., 2024). These findings imply the need for critically coordinated surveillance programs and a holistic approach to disease management on the Volta Lake other than tackling each pathogen as a single entity.

Because pathogen attribution in this survey largely relied on farmer reports and recognition of clinical signs, some misclassification is likely; such misclassification may be differential by education level and district given variation in diagnostic practices. Pathogen-specific interpretations were therefore made cautiously.

4.4 Source of fingerlings and stocking density

The findings of this study present the diversity of fingerling sources for aquaculture production on the Volta Lake, with farmers relying on their hatcheries, those of neighboring farms, or government-accredited hatcheries. The number of farmers (39%) who did not disclose information on the source of fingerling may likely reflect some farmers' poor record-keeping or mere reluctance to disclose sensitive operational details, as it may not conform with regulations for aquaculture production on the lake. The reliance on non-accredited or private sources, particularly those from neighboring farms, could pose a potential risk of horizontal transmission of pathogens through the movement of fingerlings, especially in situations where there is a lack of proper health certification and/or inadequate biosecurity measures. A rigorous certification program that helps track movement of fish across the country would be a great asset in helping authorities identify sources of diseases and help manage disease outbreaks.

Formal health certificates accompanying fingerling consignments were not consistently available and were not captured as a structured variable in this survey; instead, acquisition of health information on fingerlings and conduct of personal health assessment were recorded as proxies for pre-stocking health assurance. This finding underscores the policy relevance of standardized fingerling health certification and basic traceability mechanisms to reduce horizontal transmission risks linked to fish movements and inter-farm sourcing.

Cage dimensions and stocking density varied significantly among farmers, with some farms utilizing cages that were either significantly smaller or larger and stocking densities that ranged from very low (7 fish/m^3) to very high (>130 fish/m^3) compared to standard densities of 40–50 fish/m^3 for optimum survival and growth (Chakraborty et al., 2010; Kunda et al., 2021). Despite the wide variation in stocking density, disease incidence on the lake showed no relationship with stocking density, as the farms had diseases accompanied by mortalities, regardless of stocking density. This finding contrasts with the majority of studies on fish health and stocking densities. Low stocking densities in Nile tilapia cages have reduced risks of disease incidence (Garcia et al., 2013) compared to high stocking densities in rainbow trout that causes chronic stress, which consequently altered the hematological and immunological parameters of the fish. Additionally, high stocking density tends to negatively impact the immune response of Nile tilapia as experimental fish had low serum immunoglobulin (IgM and IgG) levels (Tammam et al., 2020). The lack of relationship between stocking density discovered in this study could be attributed to factors such as the contribution of other growth and fish health parameters, including water quality and management practices, on the various farms.

4.5 Farm management practices

This study revealed that there were variations in the management practices of cage farmers on the Volta Lake, especially regarding the grading of fish, biosecurity, water quality monitoring and disease control measures, which were critical for optimal fish health, growth and general farm productivity (Opiyo et al., 2018; Kyule-Muendo et al., 2022). Grading was found to be a common practice among the majority (62.5%) of fish farms on the lake, which is recommendable considering the adverse effects of uneven fish sizes, such as aggressiveness, cannibalism, and competition for feed, in a production system. These acts of aggression and competition can cause injuries in fish, thereby increasing their susceptibility to pathogen attacks. Despite this, the inconsistencies in the grading times before stocking or sometimes (weeks or months) post-stocking depict the lack of technical guidance or standard protocols for fish grading. Grading only once or at irregular intervals in production can result in increased fish stress, which can negatively impact fish health.

The biosecurity measures that most farmers implemented were designed to ensure they received disease-free fish stocks and to adhere to disinfection protocols. However, these measures were more informal and unstandardized. For instance, in ensuring disease-free fish stocks, most farms lacked robust quarantine protocols or actual veterinary certification. Still, they relied on visual inspection or the words of assurance from the hatchery operators. Inadequate biosecurity measures have been identified as a significant contributing factor to disease transmission, particularly in open systems such as cages (Zornu et al., 2023). Additionally, farmers are seeking more robust regulations regarding fingerling production and introductions as a means of disease management and control (Duodu et al., 2022). This is particularly important in an open system where the actions of others might have negative

consequences on a farmer's production even if such a farmer adhered to all regulations. Therefore, the lack of standardized biosecurity measures on the understudied farms may have contributed to the increased disease incidence in the lake.

The findings of this study revealed that 73.2% of farmers lacked water quality monitoring protocols. This is quite alarming, considering the effect of water quality on fish health management and productivity. Farmers' lack of a standardized water quality monitoring protocol was primarily due to logistical and financial constraints, with many citing the inability to access analytical laboratories and monitoring equipment as major obstacles. Along with these challenges, some farmers were skeptical about the effectiveness of monitoring initiatives, arguing that since the lake is an open and interconnected system, monitoring at the farm level would not result in appreciable enhancements in the overall quality of the water. This view emphasizes the necessity for integrated, basin-wide or community-level water quality management plans, where group efforts will be prioritized. Generally, though most farmers did not monitor water quality, they noticed changes in water color before and during disease outbreaks which they believe is because of discharges from nearby companies and industries. This pollution has the tendency of significantly impacting fish health in the various farms as well as the wild stocks in the lake (Turner et al., 2023).

Regarding disease control and treatment, the study discovered a dependence on non-recommended and potentially harmful treatment methods. The use of human antibiotics raises concerns about the risk of antimicrobial resistance and residue build-up, which further affects fish food safety on the lake (Dandi et al., 2024). The reported use of human antibiotics in aquaculture is a One Health concern because it may contribute to antimicrobial resistance (AMR) selection in aquatic environments and along the food chain. International guidance emphasizes prudent use of medically important antimicrobials in food-producing systems and prioritizes alternatives such as diagnostics, vaccination, and biosecurity over empiric antimicrobial use (Tang et al., 2017). Ghana's national AMR planning framework further underscores the need for coordinated surveillance and stewardship across human, animal, and environmental sectors (Anonymous, 2017). In this context, strengthening access to affordable diagnostics, improving extension/veterinary support, and implementing practical biosecurity and traceability measures may reduce reliance on antimicrobials while avoiding stigmatization of farmers operating under resource constraints.

Additionally, there were reports of conventional measures such as feed withdrawal, heat shock therapy, and the use of potassium permanganate. Although these methods may provide short-term relief from disease effects, their long-term effectiveness and safety remain uncertain without scientific proof and veterinary supervision. However, the participation of some farmers in the government-led vaccination program is recommendable as the safety and efficacy of those vaccines have been tested.

Stocking densities varied widely across farms (as reported by farmers), yet we did not observe strong evidence of an association with reported disease occurrence/mortality. This null finding should be interpreted cautiously because stocking density is likely correlated with multiple factors (farm size, fingerling sourcing, grading frequency, water quality management, and district-level conditions), and both exposure (density) and outcomes (disease attribution/mortality) are subject to measurement error. In open-water cage systems, hydrodynamics and co-varying practices may further obscure simple bivariate relationships, and the sample size limits power to detect modest effects.

Few respondents reported explicit numeric thresholds (e.g., dissolved oxygen or temperature cut-offs), suggesting that "monitoring" often reflected qualitative observation rather than instrument-based measurement. Reported barriers included feasibility constraints typical of small-to-medium cage operations (instrument cost, calibration/maintenance, training, and time). Where feasible, future studies should pair farmer reports with simple standardized measurements (handheld DO/temperature meters, turbidity/Secchi depth) and/or contextual environmental datasets (e.g., station logs or remote sensing for surface temperature/turbidity) to better align perceived upstream/downstream sources with lake circulation patterns.

4.6 Effects of farm management practices on disease-related fish death

Results from this study demonstrated that the selected management practices did not significantly predict disease-induced mortality rates within the studied population. The outcome of this analysis implies that though stocking density are known to impact fish health and general aquaculture productivity (Yarahmadi et al., 2015; Long et al., 2019; Abarike et al., 2022; Zornu et al., 2023), the impact may be influenced by other confounding factors which may not have been accounted for in this model. These may include pathogen load, feed quality, genetic variability of fish stock, or differences in the actual implementation of the reported practices. Furthermore, the reliance on farmer-reported data and the lack of proper record-keeping may introduce variability or reporting bias, which could potentially limit the accuracy of the findings. Estimates of stocking density were imprecise due to the modest analytic sample size and potential measurement error in derived density (self-reported cage dimensions and stocking numbers recorded at interview). Additionally, the issue of the lake being an open resource presents challenges to farm-level control of these management practices, which could impact the rate of disease-related mortalities.

4.7 Limitations of the study

Despite the insights gained in this study, limitations must be acknowledged. First, the non-participation of certain farmers and the lack of adjustments, largely due to logistical accessibility and privacy concerns, introduces the potential for selection bias. Also,

the reliance on farmers' recall memory, especially for mortality data, is another source of potential bias in the data collection. However, the findings of this research provide the necessary start point for future research using a more rigorous sampling approach. The final sample of 56 farms (28.6% response rate) presents a potential selection bias, since participation was based on farm accessibility and the owners' willingness to disclose information required. This means that there is the possibility of bias toward more accessible farms or farms with transparent record-keeping habits, consequently impacting the full representation of general lake disease incidence and management.

The findings indicate substantial potential for outcome and pathogen misclassification, particularly for pathogen-specific attributions derived from clinical signs and recall. Because diagnostic practices varied by district and education, misclassification may be differential, potentially inflating or attenuating apparent district-level differences and biasing pathogen-specific burden estimates in unknown directions. Accordingly, we refer to pathogen categories as "farmer-attributed" or "suspected" throughout and interpret pathogen-specific burdens cautiously.

Because mortality reporting was incomplete and the analysis was restricted to farms with reported mortality, estimates may be affected by selection related to reporting behavior; results should therefore be interpreted as associations among reporting farms rather than the full surveyed population.

4.8 Conclusion

The study confirmed that diseases continue to be a major challenge hindering the aquaculture productivity on the Volta Lake in Ghana. Diseases in the lake, according to farmers and local veterinarians, are of infectious origin, with *Streptococcus agalactiae* and ISKNV being the main causative organisms, as well as co-infections with other pathogens, such as *Aeromonas* spp. Diseased fishes show various clinical signs of bacterial or viral infections or both. Disease-related mortalities also varied among the farms, reaching 80% of stocked fish.

Despite farmers acknowledging the role of management practices, such as biosecurity measures, water quality, and stocking density, in disease occurrence and spread, the implementation of these practices on various farms was inconsistent or unstandardized. The results suggest that, although these practices are essential for fish health and disease management, farmers' isolated and inconsistent application observed in this study may have resulted in their inability to sufficiently predict disease-related mortalities, particularly in an open system like the Volta Lake. Additionally, the poor record-keeping of farmers made it difficult for them to provide accurate and adequate information on disease and mortality data during disease outbreaks.

To effectively manage and control diseases on the Volta Lake, a holistic approach is necessary among all stakeholders in the aquaculture industry in the area. Thus, farmers are expected to receive the necessary training to understand the dynamics of disease management on their various farms and the lake. It is also necessary to provide effective extension services or support to farmers, as well as well-equipped diagnostic centers around the lake to ensure rapid and more effective response to disease outbreaks. Management on the lake must follow a much more different approach than what is in place for pond farmers.

Data availability statement

The raw data supporting the conclusions of this article will be made available by the authors, without undue reservation.

Author contributions

DB: Data curation, Investigation, Methodology, Writing – original draft, Writing – review & editing. RE: Conceptualization, Data curation, Project administration, Supervision, Writing – review & editing. AG: Formal analysis, Investigation, Methodology, Supervision, Writing – review & editing. KO: Investigation, Methodology, Supervision, Writing – review & editing. ØE: Conceptualization, Data curation, Funding acquisition, Software, Visualization, Writing – review & editing, Project administration.

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Supplementary material

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